

NSAID Use Increases the Risk of Developing Hyponatremia during an Ironman Triathlon

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ABSTRACT

WHARAM P. C., D. B. SPEEDY, T. D. NOAKES, J. M. D. THOMPSON, S. A. REID, and L.-M. HOLTZHAUSEN. NSAID Use Increases the Risk of Developing Hyponatremia during an Ironman Triathlon. *Med. Sci. Sports Exerc.*, Vol. 38, No. 4, pp. 618–622, 2006. **Purpose:** Exertional hyponatremia ($\text{Na} < 135 \text{ mmol}\cdot\text{L}^{-1}$) is a potentially serious condition associated with endurance sports. It has been postulated that nonsteroidal antiinflammatory drug (NSAID) use may be a risk factor. This observational cohort study aimed to determine whether NSAID use is a risk factor for exertional hyponatremia and altered renal function during endurance exercise. **Methods:** A total of 330 athletes in the 2004 New Zealand Ironman triathlon (3.8-km swim, 180-km cycle, and 42.2-km run) were weighed before and after the race. A blood sample was drawn for measurement of plasma sodium (Na), potassium (K), urea (urea), and creatinine (creatinine) concentrations postrace. **Results:** The incidence of NSAID use was 30%, whereas the overall incidence of hyponatremia was 1.8%. NSAID use was related to the incidence of hyponatremia ($P = 0.0002$). The NSAID group had lower plasma Na ($P = 0.02$) and higher plasma K ($P = 0.002$), urea ($P = 0.05$), and creatinine ($P = 0.01$). Lower Na was also significantly related to female gender, lower prerace body weight, younger age and a smaller weight loss during the race. Race times were not associated with plasma Na; however, faster triathletes lost more weight. Estimated fluid intake was not different in the NSAID group, but heavier triathletes reported greater fluid intakes. **Conclusions:** NSAIDs are commonly used by athletes competing in endurance events and are a risk factor for hyponatremia and altered renal function. Notwithstanding high rates of NSAID use, the incidence of hyponatremia was low. We attribute this to changes in fluid replacement guidelines and drink station availability that reduce the risk of overdrinking, the principal cause of this condition. **Key Words:** RENAL FUNCTION, ENDURANCE EVENTS, FLUID BALANCE, FLUID GUIDELINES

Exercise-associated hyponatremia ($\text{Na} < 135 \text{ mmol}\cdot\text{L}^{-1}$) is a potentially serious condition reported initially in ultraendurance athletes (21) and more recently in marathon runners (3,7,11). The published data show that the etiology of symptomatic hyponatremia is fluid overload from sustained overdrinking (20). Increased occurrence of the condition was observed after publication of a position statement advising athletes to drink as much as possible during exercise (5). The first reported case in 1981 was of a collapsed

female ultraendurance runner (21). Since then, more instances have been reported in ultraendurance events (12,13,22,26,29,32), marathons (3,7,11), military training (10), and among recreational hikers (4). Asymptomatic hyponatremia is identified only by blood testing. Symptomatic hyponatremia may present with vomiting, dyspnea, altered levels of consciousness, and convulsions due to hyponatremic encephalopathy (32). Death due to hyponatremia has been reported following ultraendurance events (12,29), 42-km marathons (3), and military training (10).

Why the kidneys do not excrete the excessive fluid load in these athletes is unknown. When studied at rest, renal response to a fluid challenge was not different between triathletes who had previously developed hyponatremia during an Ironman triathlon and those who had finished the same race without developing hyponatremia (27).

Risk factors associated with exertional hyponatremia are female gender and slower race times (7,11,29). Nonsteroidal antiinflammatory drug (NSAID) use may also be a risk factor, although the data are conflicting (3,7,11) and

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TABLE 1. Biochemical parameters ($\text{mmol}\cdot\text{L}^{-1}$) for triathletes who either did (NSAID) or did not (No NSAID) ingest nonsteroidal antiinflammatory drugs during the 2004 New Zealand Ironman Triathlon.

Measure	NSAID (<i>N</i> = 100)	No NSAID (<i>N</i> = 233)	<i>T</i> Value (<i>P</i> Value)
Plasma Na	140.2 (3.36)	141.1 (3.07)	2.35 (0.02)
Plasma K	4.51 (0.46)	4.35 (0.43)	-3.14 (0.002)
Plasma urea	9.72 (2.59)	9.15 (1.88)	-1.97 (0.05)
Plasma creatinine	0.13 (0.03)	0.12 (0.02)	-2.54 (0.01)

Values are reported as mean (SD).

no prospective study has yet investigated this hypothesis. Under resting conditions, NSAIDs can cause hyponatremia by reducing renal free water clearance, although this is rare (23). NSAIDs inhibit renal prostaglandin synthesis (8) and may impair kidney function (17). Renal perfusion is maintained during exercise by vasodilatory prostaglandins, despite the reduction in renal blood flow (35). Indeed, reduced glomerular filtration rate during exercise is associated with NSAID use (9). Furthermore, a recent study has shown altered renal function in NSAID-medicated runners completing a standard 42-km marathon (24). This raises the possibility that NSAIDs may cause or increase the risk of exertional hyponatremia in athletes who overconsume fluids. Both cyclooxygenase-1 (COX-1) and COX-2-related prostaglandins are expressed in the kidney and NSAIDs and COX-2 selective inhibitors show similar effects on renal function (36). Many athletes use NSAIDs during endurance events (14).

This research project aimed therefore to, first, determine the incidence of NSAID use at an ultradistance Ironman triathlon and, second, to determine whether NSAIDs are a risk factor for the development of exertional hyponatremia under these conditions. We hypothesized that NSAIDs would impair prostaglandin synthesis during exercise thereby decreasing free water clearance with resulting fluid retention.

METHODS

Approval for the study was obtained from the local ethics committee. All 1339 entrants at the 2004 New Zealand Ironman ultradistance triathlon were invited to participate in this study. Written informed consent was obtained from 388 triathletes. Blood samples were collected from 333 triathletes (264 males and 69 females) at the completion of the event. Postrace weights were not recorded three subjects. As a result, analysis of the relationship of blood parameters to weight changes was performed on 330 triathletes. Of the 330 137 (42%) triathletes were able to estimate their fluid intakes during the race.

Each triathlete swam 3.8 km, cycled 180 km, and ran 42.2 km consecutively. The race began at 7:00 a.m. Ambient air temperature at 12:00 p.m. on the race day was 14.5°C with a relative humidity of 74%. Water temperature was 15.8°C. Food and drink were freely available at support stations situated every 18 km on the cycle course and every 2.4 km on the run. Fluids available included water, Coca-Cola®, and a sport drink (Pro4®) containing 20 $\text{mmol}\cdot\text{L}^{-1}$ of sodium.

All triathletes were weighed at race registration and then reweighed at the finish line in their running clothes and shoes. Race times for swim, cycle, and run were recorded.

Blood was collected by routine venipuncture from all consenting triathletes within 30 min of their finishing the race. Triathletes either sat or lay supine during venipuncture. Blood was collected in heparinized silicone-gel separator tubes, immediately centrifuged for 20 min, and then stored at room temperature. Heparinized samples were collected for plasma analysis to avoid fibrin clotting and artificial elevation of potassium concentrations. Assays were performed the next day with a Hitachi 917 analyzer (Boehringer, Mannheim), using standard methods and the manufacturer's reagents. Blood was analyzed for plasma Na, K, chloride, urea, and creatinine concentrations. For the purpose of this study, hyponatremia was defined as plasma Na < 135 $\text{mmol}\cdot\text{L}^{-1}$.

Use of NSAIDs in the 24 h prior to the race finish was recorded. Fluid intake during the race was recorded via a questionnaire at the time of venipuncture from which the intake of water, Coca-Cola®, and the Pro4® sports drink during the race was estimated.

Statistical analysis. The data were analyzed using Student's *t*-tests to identify differences between groups and linear regression to identify the relationships with continuous variables. χ^2 statistics were used to analyze 2×2 tables. Multivariable analysis controlling for potential confounding variables was performed using multiple linear regression. A value of $P < 0.05$ was regarded as significant for all determinations.

RESULTS

The overall incidence of NSAID use was 30% (100/333). There were no hyponatremic triathletes in the non-NSAID group, whereas six triathletes in the NSAID group finished the race with Na < 135 $\text{mmol}\cdot\text{L}^{-1}$; this was statistically significant ($\chi^2 = 14.24$, $P = 0.0002$). The incidence of hyponatremia was therefore 1.8% (6/333), but 6% in the NSAID group. All hyponatremic triathletes were asymptomatic, and none had a plasma Na concentration < 130 $\text{mmol}\cdot\text{L}^{-1}$.

Plasma Na was significantly lower in the NSAID group, whereas plasma K, urea, and creatinine were all significantly higher in the NSAID group (Table 1). However, differences in mean values are small and unlikely to be of clinical significance. Decreased plasma Na was significantly related to female gender, NSAID use, lower prerace

TABLE 2. Factors associated with alterations in plasma Na.

	Univariate (95% CI)	Multivariate (95% CI)
Female	-1.88 (-2.70 to -1.06)	-0.55 (-1.51 to 0.40)
NSAID use	-0.89 (-1.63 to -0.15)	-0.88 (-1.54 to -0.21)
Prerace weight (per kilogram)	0.073 (0.041 to 0.106)	0.048 (0.011 to 0.085)
Age	0.06 (0.02 to 0.10)	0.07 (0.04 to 0.11)
Weight loss (%)	-0.61 (-0.78 to -0.44)	-0.60 (-0.76 to -0.43)

Table shows parameter estimates (slope of line) and 95% confidence intervals (CI) for the estimates. Parameter estimates for continuous variables relate to a one-unit change.

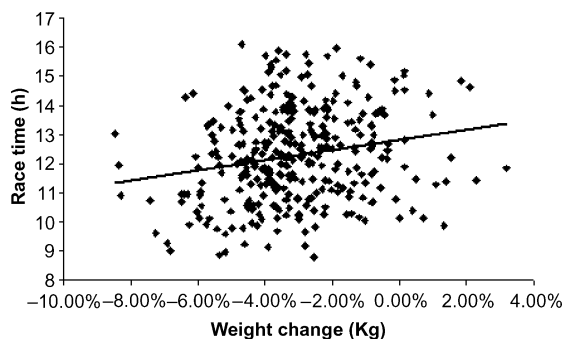


FIGURE 1—Faster finishing times are associated with higher levels of weight loss during an Ironman triathlon.

weight, younger age, and lesser weight loss but not race time. When these variables were controlled for in multivariable analysis, NSAID use, lower prerace weight, lesser weight loss, and age remained statistically significant (Table 2). Greater relative weight loss was associated with faster finishing times, equivalent to a 1-h reduction in race finish time for each 0.2% weight loss during the race ($P = 0.0004$) (Fig. 1).

No differences were found in the characteristics of those who provided information on their fluid intakes during the race compared with the whole study group. Related to fluid intake no association was found with age, overall time (or any individual component), or with NSAID use. However, increased fluid intake was significantly related to both a higher prerace weight (equivalent to $56\text{-mL}\cdot\text{kg}^{-1}$ increase in weight, $P = 0.0003$) (Fig. 2) and male gender ($P = 0.05$). Controlling for both of these variables in multivariable analysis revealed that only prerace weight was still a significant predictor of total fluid intake during the race ($P = 0.002$).

Relative weight change was significantly related to fluid intake (0.17% increase in relative weight change per milliliter of fluid intake), and this continued to be significant when controlling for prerace weight (0.16% increase in weight per milliliter of fluid intake). This means an athlete will gain 1% in weight during the race by drinking 6.1 L of fluid.

DISCUSSION

This study is the first to show that NSAID use is associated with an increased incidence of hyponatremia in Ironman triathletes. The association between exertional hyponatremia and NSAID use is highly significant ($P = 0.0002$), confirming the postulate that NSAIDs are a risk factor for hyponatremia (7). All six hyponatremic subjects had taken NSAIDs; with the potential for serious medical complications from hyponatremia, the demonstration that NSAIDs are a risk factor is important. Plasma Na was significantly lower in the NSAID group and NSAID use was also an independent risk factor for lower plasma Na postrace. As estimated fluid intake was not different in the NSAID group, the lower plasma Na may reflect altered

renal function resulting in increased fluid retention despite equivalent rates of fluid intake. The use of NSAIDs by 30% of competitors in the New Zealand Ironman concurs with previous studies reporting high use of these medications by endurance athletes (11).

The second important finding of our study is that NSAIDs appear to effect renal function during exercise in the race situation. Plasma creatinine, urea, and K concentrations were all significantly increased in the NSAID group (Table 1). This confirms a previous study that found altered renal function in runners completing a standard distance marathon after taking NSAIDs (24). The prostaglandins expressed in the kidney are involved with the control of renal hemodynamics and glomerular filtration as well as salt and water filtration (36). Although we hypothesized that NSAIDs may be a risk factor for the development of hyponatremia due to reduced glomerular filtration rate, higher postrace body weight and NSAID use were independent risk factors for lower plasma Na postrace. NSAIDs have also been shown to affect Na-K exchange in the distal renal tubules; this mechanism may partly explain the altered renal function observed in the NSAID group (33). NSAIDs and COX-2 selective inhibitors show similar effects on renal function (36). We did not identify NSAIDs by individual drug or COX-2 selective inhibition. This precluded separate analysis of the data, which would require further investigation to assess whether NSAIDs and COX-2 selective inhibitors have similar effects during exercise. Hyponatremia reported with NSAID use by nonathletes has been suggested to result from inappropriate arginine-vasopressin (AVP) secretion (25); however, a study of Ironman triathletes revealed normal AVP concentrations among hyponatremic participants (31).

A further observation of this study is the finding that lighter prerace body weights are also independently associated with lower plasma Na postrace. Montain et al. (16) have postulated that smaller individuals require a lesser degree of fluid overload to dilute their extracellular Na, compared to larger individuals. Indeed, guidelines that stress rigid fluid intake rates may promote rates of fluid intake that exceed the tolerance of lighter athletes (2,4).

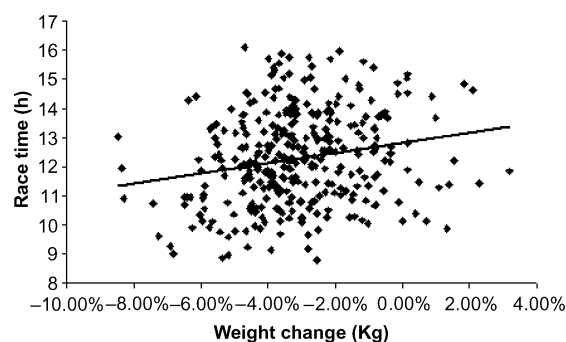


FIGURE 2—Linear relationship between prerace weight and total fluid during an Ironman triathlon.

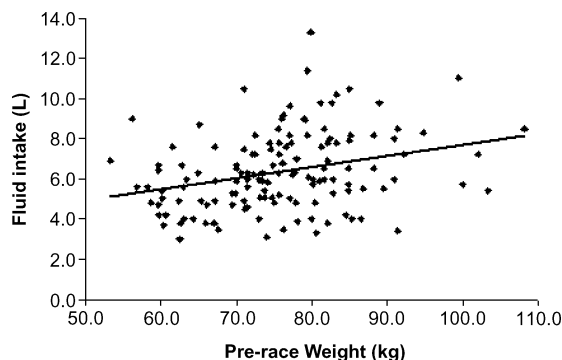


FIGURE 3—Inverse relationship between postrace plasma Na and change in body weight in 330 Ironman triathletes.

Female gender was also a highly significant risk factor for lower plasma Na ($P = 0.0001$), confirming previously reported findings (29). After controlling for other variables in the multivariate analysis, the effect of female gender was no longer significant (Table 2). The effect of female gender is accounted for by lower initial body weights and lesser relative weight loss during the race. Slower race times (and therefore more opportunity to consume fluid) have been suggested as the reason for the increased risk in female athletes (5,16). It is perhaps safest to recommend that athletes drink *ad libitum* (according to the dictates of thirst) (20) with an upper limit of no more than 400–800 mL·h⁻¹ (the lower limit for slower lighter athletes) (19). Age also had a small effect, as younger athletes were more likely to have lower postrace plasma Na.

Weight loss at the end of the race was related to faster finishing times (Fig. 1) in agreement with similar findings at the South African Ironman (26). This confirms Noakes' (18) observations that faster runners are likely to be more dehydrated and that this does not necessarily affect their performance in the field (6). Laboratory-based studies have suggested reduced performance is associated with dehydration (15), but this finding has been difficult to replicate in field studies (19). No association between plasma Na and race times was noted. Previous studies have found an association between low plasma Na and slower race times, but the low incidence of hyponatremia in the current study may explain our failure to observe this relationship (34).

REFERENCES

1. ALMOND, C. S., E. B. FORTESCUE, A. Y. SHIN, R. MANNIX, and D. S. GREENES. Risk factors for hyponatremia among runners in the Boston marathon. *Acad. Emerg. Med.* 10:534–535, 2003.
2. ARMSTRONG, L. E., C. M. MARESH, C. V. GABAREE, et al. Thermal and circulatory responses during exercise: effects of hypohydration, dehydration and water intake. *J. Appl. Physiol.* 82: 2028–2035, 1997.
3. AYUS, J. C., J. VARON, and A. I. ARIEFF. Hyponatremia, cerebral edema, and noncardiogenic pulmonary edema in marathon runners. *Ann. Intern. Med.* 132:711–714, 2000.
4. BACKER, H. D., E. SHOPE, S. L. COLLINS, and H. BARKAN. Exertional heat illness and hyponatremia in hikers. *Am. J. Emerg. Med.* 17: 532–539, 1999.
5. CONVERTINO, V. A., L. E. ARMSTRONG, and E. F. COYLE. American College of Sports Medicine position stand: exercise and fluid replacement. *Med. Sci. Sports Exerc.* 28:i–vii, 1996.
6. DARIES, H. N., T. D. NOAKES, and S. C. DENNIS. Effect of fluid intake volume on 2-h running performances in a 25 degree C environment. *Med. Sci. Sport Exerc.* 32:1783–1789, 2000.
7. DAVIS, D. P., J. S. VIDEEN, A. MARINO, et al. Exercise-associated hyponatremia in marathon runners: a two-year experience. *J. Emerg. Med.* 21:47–57, 2001.
8. DELMAS, P. D. Non-steroidal anti-inflammatory drugs and renal function. *Br. J. Rheumatol.* 34:25–28, 1995.
9. FARQUHAR, W. B., A. L. MORGAN, E. J. ZAMBRASKI, and W. L. KENNEY. Effects of acetaminophen and ibuprofen on the

It has been estimated that 2.5 kg in weight is lost from glycogen stores and metabolic water during an Ironman event (28). None of the hyponatremic athletes in this study group lost more than 2.5 kg, which conflicts with the concept of “dehydration hyponatremia” (2,12) and is consistent with published data showing that hyponatremia is due to fluid overload (Fig. 3) (13).

An important finding of this study was the very low incidence of hyponatremia at the 2004 New Zealand Ironman triathlon (1.8%). This compares to an incidence of 18% in the same race in 1997 (29). Indeed, symptomatic exercise-associated hyponatremia is almost an historical disease in New Zealand (Dr. Lucy Holtzhausen, personal communication). This has been attributed to a reduction in the number of support stations (thus reducing fluid availability) and education regarding appropriate fluid intake (30). In an Ironman triathlon in South Africa, where these same education and fluid availability strategies were followed, hyponatremia is also very uncommon and present only in those who fail to adopt these practices (22). This contrasts with recent North American marathons where incidences of hyponatremia of 9–13% have been reported (1,11). We believe that, with improved understanding and education related to appropriate fluid intake during ultraendurance competitions, symptomatic hyponatremia will become a historical entity at these events.

CONCLUSIONS

This study shows that NSAIDs are commonly used in ultraendurance events and that their use is associated with an increased risk of the development of biochemically diagnosed exertional hyponatremia. This effect is likely to be due to an alteration in renal function. These are important findings, and athletes, health practitioners, and pharmacists need to be informed accordingly. The very low incidence of hyponatremia at the 2004 New Zealand Ironman event, despite a high incidence in the past, suggests that more recent fluid intake guidelines and changes to fluid availability at this event have largely eliminated this serious condition.

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- renal function of the stressed kidney. *J. Appl. Physiol.* 86: 598–604, 1999.
10. GARIGAN, T. P., and D. E. RISTEDT. Case report: death from hyponatremia as a result of acute water intoxicification in an army basic trainee. *Milit. Med.* 164:234–238, 1999.
 11. HEW, T. D., J. N. CHORLEY, J. C. CIANCA, and J. G. DIVINE. The incidence, risk factors, and clinical manifestations of hyponatremia in marathon runners. *Clin. J. Sports Med.* 13:41–47, 2003.
 12. HILLIER, W. D. B. Dehydration and hyponatremia during triathlons. *Med. Sci. Sports Exerc.* 21:219–221, 1989.
 13. IRVING, R. A., T. D. NOAKES, R. BUCK, et al. Evaluation of renal function and fluid homeostasis during recovery from exercise-induced hyponatremia. *J. Appl. Physiol.* 70:342–348, 1991.
 14. KENNEY, W. L. Nonsteroidal anti-inflammatory drugs and the kidney. *Curr. Sports Med. Rep.* 1:125–126, 2002.
 15. LATZKA, W. A., and S. J. MONTAIN. Water and electrolyte requirements for exercise. *Clin. J. Sports Med.* 18:513–524, 1999.
 16. MONTAIN, S. J., M. N. SAWKA, and C. B. WENGER. Hyponatremia associated with exercise: risk factors and pathogenesis. *Exerc. Sports Sci. Rev.* 29:113–117, 2001.
 17. MURRAY, M. D., and D. C. BRATER. Renal toxicity of the nonsteroidal anti-inflammatory drugs. *Annu. Rev. Pharmacol. Toxicol.* 33:435–465, 1993.
 18. NOAKES, T. D. Dehydration during exercise: what are the real dangers? *Clin. J. Sport Med.* 5:123–128, 1995.
 19. NOAKES, T. D. Fluid replacement during marathon running. *Clin. J. Sports Med.* 13:309–318, 2003.
 20. NOAKES, T. D. Hyponatremia in distance runners: fluid and sodium balance during exercise. *Curr. Sports Med. Rep.* 4:197–207, 2002.
 21. NOAKES, T. D., N. GOODWIN, B. L. RAYNER, T. BRANKEN, and R. K. TAYLOR. Water intoxication: a possible complication during endurance exercise. *Med. Sci. Sports Exerc.* 17:370–375, 1985.
 22. NOAKES, T. D., K. SHARWOOD, M. COLLINS, and D. R. PERKINS. The dipsomania of great distance: water intoxication in an Ironman triathlete. *Br. J. Sports Med.* 38:e16, 2004.
 23. PETERSSON, I., G. NILSSON, B. G. HANSSON, and T. HEDNER. Water intoxication associated with non-steroidal anti-inflammatory drug therapy. *Acta Med. Scand.* 221:221–223, 1987.
 24. REID, S. A., D. B. SPEEDY, J. M. THOMPSON, et al. Study of hematological and biochemical parameters in runners completing a standard marathon. *Clin. J. Sport Med.* 14:344–353, 2004.
 25. SANDERS, B., T. D. NOAKES, and S. C. DENNIS. Sodium replacement and fluid shifts during prolonged exercise in humans. *Eur. J. Appl. Physiol.* 84:419–425, 2001.
 26. SHARWOOD, K., M. COLLINS, J. GOEDECKE, G. WILSON, and T. D. NOAKES. Weight changes, medical complications, and performance during an Ironman triathlon. *Br. J. Sports Med.* 38:718–724, 2004.
 27. SPEEDY, D. B., T. D. NOAKES, T. BOSWELL, J. M. THOMPSON, N. REHRER, and D. R. BOSWELL. Response to a fluid load in athletes with a history of exercise induced hyponatremia. *Med. Sci. Sports Exerc.* 33:1434–1442, 2001.
 28. SPEEDY, D. B., T. D. NOAKES, N. E. KIMBER, et al. Fluid balance during and after an Ironman triathlon. *Clin. J. Sport Med.* 11: 44–50, 2001.
 29. SPEEDY, D. B., T. D. NOAKES, I. R. ROGERS, et al. Hyponatremia in ultradistance triathletes. *Med. Sci. Sports Exerc.* 31:809–815, 1999.
 30. SPEEDY, D. B., I. R. ROGERS, T. D. NOAKES, et al. Diagnosis and prevention of hyponatremia at an ultradistance triathlon. *Clin. J. Sports Med.* 10:52–58, 2000.
 31. SPEEDY, D. B., I. R. ROGERS, T. D. NOAKES, et al. Exercise-induced hyponatremia in ultradistance triathletes is caused by inappropriate fluid retention. *Clin. J. Sports Med.* 10:272–278, 2000.
 32. SPEEDY, D. B., I. R. ROGERS, S. SAFI, and B. FOLEY. Hyponatremia and seizures in an ultradistance triathlete. *J. Emerg. Med.* 18:41–44, 2000.
 33. STRUMER, T., M. M. ELSEVIERS, and M. E. DE BROE. Nonsteroidal anti-inflammatory drugs and the kidney. *Curr. Opin. Nephrol. Hypertens.* 10:161–163, 2001.
 34. VRIJENS, D. M., and N. J. REHRER. Sodium-free fluid ingestion decreases plasma sodium during exercise in the heat. *J. Appl. Physiol.* 86:1847–1851, 1999.
 35. WHELTON, A. Nephrotoxicity of nonsteroidal anti-inflammatory drugs: physiologic foundations and clinical implications. *Am. J. Med.* 106:13S–24S, 1999.
 36. WHELTON, A. Renal aspects of treatment with conventional nonsteroidal anti-inflammatory drugs versus cyclooxygenase-2-specific inhibitors. *Am. J. Med.* 110:33S–42S, 2001.